

DSOLVE 2026 – Candidate Guidelines & Event Brochure

DRISHTI | College of Engineering Trivandrum (CET)

BUILD. SOLVE. DEMONSTRATE.

- **Format:** 36–Hour Physical Hackathon
- **Venue:** College of Engineering Trivandrum (CET)
- **Team Size:** 2 to 4 Members (Same Institution)
- **Focus Area:** Dental Practice & Health Tech Innovation

1. About DSOLVE 2026

DSOLVE is an intensive **36–hour physical hackathon** conducted as part of DRISHTI at the College of Engineering Trivandrum (CET). Participants collaborate to build innovative, practical, and technology–driven solutions addressing real-world challenges in the dental practice and health tech industry.

2. Eligibility & Team Guidelines

- **Open Eligibility:** Open to all university and college students across all academic branches, years, and institutions.
- **Team Composition:** Teams must consist of **2 to 4 members**.
- **Institutional Rule:** All team members **must belong to the same college/institution**. Inter-college teams are strictly not permitted.
- **Single Registration:** Each candidate may register only once and participate in exactly one team. Duplicate registrations or multi-team participation will result in immediate disqualification.
- **Roster Lock:** Team members cannot be changed after registration closes.

3. Problem Statements

Problem 1: Oral Health Screening Widget

Develop a free, two-minute oral health screening widget for web or smartphones that guides patients through a simple set of prompts and captures five quick images of their teeth.

The solution should analyse these images and generate an instant visual report highlighting potential oral health concerns such as crooked teeth, tooth wear, or discoloration. The goal is to provide patients with an easy, accessible way to get an initial visual assessment of their oral health and understand whether they may need to consult a dentist.

Problem 2: Dental Practice Marketing Automation

Develop an automated marketing solution for dental practices that can manage and act on patient data collected from multiple sources. The solution should focus on key workflows such as:

- Identifying and converting potential leads received through website enquiries or emails.
- Engaging existing patients who have not scheduled their next appointment and following up with them.
- Automating post-appointment communication and encouraging patients to leave reviews.

The goal is to reduce manual marketing effort for dental practices while improving patient engagement and follow-up.

Problem 3: Insurance Verification Automation

Develop a solution to simplify and automate insurance verification for dental practices.

Currently, dental practices often need to manually contact insurance companies to verify whether a patient's insurance is active and what coverage is available. This

process can take 20–30 minutes per patient, creating significant administrative effort and delays.

The solution should explore ways to automate or significantly reduce this manual process by quickly verifying insurance eligibility and presenting the relevant information to the dental practice in a simple and usable format.

Problem 4: Automated Camera Workflow

Develop an automated camera workflow that enables clinical teams to capture and transfer patient images directly from professional cameras into a dental software application.

Currently, clinical teams may need to manually capture, transfer, organise, and upload patient images, and workflows can vary across camera brands.

The solution should support commonly used professional DSLR cameras such as **Canon**, **Nikon**, and **Sony**, and enable seamless streaming or transfer of patient images directly into the application, reducing manual steps and improving the clinical photography workflow.

Problem 5: Digital Smile Design Simulation

Develop a visual digital smile design tool that allows dental professionals and patients to simulate how a patient's teeth could look after a proposed dental treatment.

The solution should enable users to visualise potential changes to the patient's smile based on specific treatment options. The goal is to make treatment outcomes easier to understand and support better communication between dentists and patients.

Problem 6: Open Problem Statement – Dental Industry

Identify and solve a real-world problem within the dental industry across markets such as the UK, Australia, and the US.

Participants are encouraged to explore challenges faced by dental practices, dentists, clinical teams, patients, or dental organisations and propose a technology-driven solution.

The problem can relate to areas such as clinical workflows, patient experience, practice operations, insurance, billing, marketing, diagnostics, communication, or automation.

The solution should clearly define the problem being addressed, demonstrate how the proposed technology can solve it, and showcase a working prototype or proof of concept.

Open problem statements must focus on the dental practice industry, specifically targeting markets like the US and UK. For inspiration and idea generation, participants are encouraged to explore platforms like G2 and Capterra, or research Y Combinator startups operating in this domain.

4. Timeline & Event Milestones

Milestone / Review	Estimated Window	Format & Scope
Reporting Time	17th Thu, 04:45 PM	Candidate check-in and arrival at venue
Hackathon Commences	17th Thu, 06:00 PM	Official start of the 36-hour hacking sprint
First Interim Check	17th Thu, ~7:00 PM	Initial check-in: ideation, architecture & tech stack

Milestone / Review	Estimated Window	Format & Scope
		setup (mentors visit team tables)
Second Interim Check	18th Fri, ~10:30 AM	Mid-sprint check-in: core functionality review, progress evaluation & mentor feedback
Third Interim Check	18th Fri, ~6:30 PM	Final review: pre-submission status check, pitch video guidance & slide deck prep
Presentation & Judging	19th Sat, 09:00 AM	Live demonstration and technical evaluation with judges

Day-by-Day Detailed Agenda

Day 1: Thursday, Sept 17th

- **04:45 PM:** Reporting Time
- **05:00 PM – 05:30 PM:** Inauguration
- **06:00 PM:** Hackathon Commences
- **08:00 PM – 09:00 PM:** Dinner
- **09:00 PM:** Hackathon Contd.

Day 2: Friday, Sept 18th

- **01:00 AM:** Informals & Icebreaking
- **02:00 AM:** Hackathon Contd.
- **08:00 AM – 09:00 AM:** Breakfast
- **09:00 AM:** Hackathon Contd.
- **12:30 PM – 01:30 PM:** Lunch

- **01:30 PM:** Hackathon Contd.
- **04:30 PM – 05:00 PM:** Tea & Snacks
- **05:00 PM:** Hackathon Contd.
- **08:00 PM – 09:00 PM:** Dinner
- **09:00 PM:** Hackathon Contd.

Day 3: Saturday, Sept 19th

- **05:00 AM:** Submission Prep Reminder
- **06:00 AM:** Code Freeze
- **08:00 AM – 09:00 AM:** Breakfast
- **09:00 AM – 12:30 PM:** Presentation & Judging (Morning)
- **12:30 PM – 01:30 PM:** Lunch
- **01:30 PM – 04:30 PM:** Presentation & Judging (Afternoon)
- **05:00 PM – 06:00 PM:** Results & Closing Ceremony

5. Presentation, Pitch Video & Judging Guidelines

- **Social Media Pitch Video:**
 - **Duration:** >30 seconds short-form video highlighting the core problem and proposed solution.
 - **Language:** Must be presented clearly in English.
 - **Production:** Video creation and editing must be completed entirely within the 36-hour hackathon timeframe.
 - **Distribution:** Must be posted on participants' personal social media profiles, tagging **Drishti** and **CareStack** prior to 6:00 AM on Sept 19th.
- **Final Presentation Format:**
 - **Live Pitch & Demo:** 3 to 5 minutes live demonstration of the functional prototype.
 - **Technical Q&A:** 5 to 10 minutes of in-depth technical and product evaluation with judges.
 - **Slide Template:** No rigid slide format enforced.

6. Development Rules, AI Policy & Submissions

- **36-Hour Core Development:** All core code must be built during the official 36-hour window. Pre-built projects or reusing substantial pre-existing solutions are strictly forbidden.
- **Permitted Resources:** Open-source frameworks, public APIs, datasets, documentation, and pre-existing AI models are fully permitted.
- **Tech Stack Flexibility:** Teams have complete freedom in tech stack selection. Teams may pivot problem statements at any time during the event.
- **AI Tool Usage:** AI coding assistants, generative models, and design tools are permitted as development aids. However, participants must be able to explain, defend, and justify all code, architecture, and design choices.
- **GitHub Repository:** Every team must maintain and submit a public GitHub repository containing clean source code, comprehensive README documentation, and setup instructions. Organizers will provide a minimal repository setup template.

7. Venue, Infrastructure & Code of Conduct

- **Physical Event Presence:** DSOLVE is an in-person event hosted at CET. Participants must remain active and available on-site for mentor checks and periodic evaluations.
- **Provided Facilities:** High-speed Wi-Fi internet, electrical power access at workstations, and designated hacking spaces will be provided.
- **Participant Requirements:** Attendees must bring their own laptops, chargers, hardware peripherals, and valid student identification cards (and a valid government ID such as Aadhaar Card, Driving License, etc...).
- **Code of Conduct:** Participants must maintain professionalism, communicate respectfully, and adhere to all campus safety guidelines throughout the event. Any damage to property or harm to others will result in strict disciplinary action and personal liability. Furthermore, plagiarism, the use of pre-built submissions, or any form of misconduct will lead to immediate disqualification.

8. Terms and Conditions

- **Finality of Decisions:** The rulings of the judges and organizers regarding rules, scoring, and awards are final, binding, and not subject to appeal.
- **Document Updates:** These guidelines are subject to change, and periodic updates may be issued by the organizing team. Participants are encouraged to review this document regularly for any changes.